

Total no. of Pages:

Roll no.

6th Semester

MID-SEMESTER EXAMINATION, FEBRUARY-MARCH, 2026

Course Title: Optimization Techniques in AI & ML

Course Code: EAEPE21

Time: 1.5 Hours

Max. Marks: 15

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	What is the utility of optimization?	1.5	CO1
1b	Write the generic form of optimization problems.	1.5	CO1
2a	What does the No Free Lunch Theorem imply ?	1.5	CO1
2b	What is a Metaheuristic Algorithm? Compare Heuristic and Metaheuristic Algorithms.	1.5	CO1
3a	What is exploration and exploitation in terms of optimization? How to maintain a balance between them?	1.5	CO2
3b	Explain simulated annealing algorithm with an example.	1.5	CO2
4a	What is Global and Local Optima? How to avoid local optima?	1.5	CO2
4b	Write the pseudo code for a standard genetic algorithm.	1.5	CO2
5a	What are the advantages of PSO algorithm compared to other traditional algorithms.	1.5	CO1
5b	Using Simulated Annealing algorithm, initialize the required parameters and minimize the function $x^2 - x + 2$ for two iterations.	1.5	CO2

Total no. of Pages... 1

SEMESTER- (BTech(CSDA) sixth Sem/ECAM 8th Sem(EG)/Backlog CSDA eighth sem-
Course Work

MID-SEMESTER EXAMINATION, Feb, 2026

Course Code-. CBCPE24

Course Title- Artificial Neural Network

Time-. 1 ½ . Hour

Max. Marks- 15.

Note: Attempt all questions. Missing data/information(if any), may be suitably assumed & mentioned in the answer.

Q.No	Question	Marks	CO
1a	Differentiate biological neural network and artificial neural network?	1	CO1
1b	Describe different activation functions.	2	CO1
2a	Explain ADALINE neural network?	1	CO1
2b	Discuss any four characteristics of neural networks that distinguish them from traditional computing systems.	2	CO1
3a	Define the error function in backpropagation.	1	CO1
3b	A McCulloch-Pitts neuron has two inputs x_1 and x_2 with weights $w_1 = 1$ and $w_2 = 1$. The threshold $\theta = 1.5$. Determine the output for all possible binary input combinations. Which logic gate does this neuron implement?	2	CO1
4a	Differentiate deep learning and machine learning.	1	CO2
4b	Explain backpropagation algorithm with an example.	2	CO2
5a	Write the use of tuning hyper parameters in ANN?	1	CO2
5b	Explain how ANN can be applied to provide employment to rural women.	2	CO2